

Norway

SLEIPNER

Volve F

F-11

15/9-F-11 B

Plan: 15/9-F-11 B IOR DG3 rev0

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B IOR DG3 rev0	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-11		
Well Position	+N/-S	4.65 m	Northing:
	+E/-W	-0.93 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.9566 N
			Longitude:
			1° 53' 14.8667 E
			Water Depth:
			91.00 m

Wellbore	15/9-F-11 B		
Magnetics	Model Name	Sample Date	Declination
			(°)
	MNETICREFERENCE	27.09.2012	-1.29
			Dip Angle
			(°)
			71.67
			Field Strength
			(nT)
			50,475

Design	15/9-F-11 B IOR DG3 rev0		
Audit Notes:			
Version:	Phase:	PLAN	Tie On Depth:
			2,570.00
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	4.65	-0.93
			Direction
			(°)
			91.28

Survey Tool Program	Date	02.09.2016		
From	To	Survey (Wellbore)	Tool Name	Description
(m)	(m)			
150.00	2,570.00	F-11 T2 SDI Keeper Gyro (15/9-F-11 T2)	Keeper, cont	Gyro Tool from SDC
2,570.00	3,200.00	15/9-F-11 B IOR DG3 rev0 (15/9-F-11 B)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
3,200.00	4,864.71	15/9-F-11 B IOR DG3 rev0 (15/9-F-11 B)	Magn, IFR, non-mag, dual in	Magnetic Tools (MWD, EMS)

Planned Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
2,570.00	36.93	68.78	2,440.31	39.36	-348.15	-347.91	0.000	0.000	0.000	
2,570.68	36.98	68.81	2,440.85	39.50	-347.77	-347.53	2.320	2.159	1.410	
2,580.68	36.98	68.81	2,448.84	41.68	-342.16	-341.97	0.000	0.000	0.000	
2,841.21	62.35	76.79	2,616.25	97.34	-153.50	-154.61	3.000	2.922	0.918	
2,927.32	62.35	76.79	2,656.20	114.77	-79.24	-80.76	0.000	0.000	0.000	
3,069.05	62.00	92.82	2,722.70	126.10	45.00	43.20	3.000	-0.074	3.394	
3,196.53	63.96	106.97	2,780.84	106.54	156.45	155.06	3.000	0.462	3.329	
3,200.00	63.96	106.97	2,782.36	105.63	159.43	158.06	0.000	0.000	0.000	
3,210.00	63.96	106.97	2,786.75	103.01	168.02	166.71	0.000	0.000	0.000	

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Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
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Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B IOR DG3 rev0	Database:	Production EDM P246N

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,302.88	65.55	117.09	2,826.44	71.51	245.74	245.11	3.000	0.512	3.269	
3,310.12	65.55	117.09	2,829.44	68.51	251.61	251.04	0.000	0.000	0.000	
3,393.10	72.70	121.60	2,859.00	30.49	319.09	319.36	3.000	2.585	1.631	
3,485.04	70.92	130.79	2,887.70	-21.03	389.52	390.92	2.906	-0.580	2.998	
3,702.68	89.09	118.56	2,925.45	-141.68	565.06	569.12	3.000	2.504	-1.686	
3,815.55	89.09	118.56	2,927.25	-195.63	664.19	669.42	0.000	0.000	0.000	
3,829.30	88.00	119.40	2,927.60	-202.29	676.21	681.59	3.000	-2.373	1.836	
3,967.85	86.54	105.61	2,934.24	-255.14	803.75	810.27	3.000	-0.317	-2.987	
4,008.31	86.54	105.61	2,936.68	-266.00	842.65	849.41	0.000	0.000	0.000	
4,137.67	79.00	95.00	2,953.00	-289.00	968.62	975.86	3.000	-1.748	-2.460	
4,306.36	65.30	85.34	3,004.70	-290.00	1,128.57	1,135.80	2.932	-2.437	-1.718	
4,506.40	71.22	64.28	3,079.50	-241.01	1,306.33	1,312.42	3.062	0.888	-3.159	
4,664.71	70.30	47.50	3,132.00	-157.61	1,429.58	1,433.77	3.005	-0.173	-3.179	
4,864.71	70.30	47.50	3,199.41	-30.40	1,568.41	1,569.73	0.000	0.000	0.000	

Design Targets										
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude	
F-11 BCU Guide Target - plan hits target center - Point	0.00	360.00	2,859.00	30.49	319.09	6,478,594.00	435,369.00	58° 26' 30.9622 N	1° 53' 34.5628 E	
F-11 Top Hugin Target - plan misses target center by 0.06m at 3485.04m MD (2887.76 TVD, -21.00 N, 389.51 E) - Point	0.00	360.00	2,887.70	-21.03	389.51	6,478,542.50	435,439.40	58° 26' 29.3349 N	1° 53' 38.9553 E	
F-11 IUH Guide 1 Target - plan hits target center - Point	0.00	360.00	2,927.60	-202.29	676.21	6,478,361.30	435,726.00	58° 26' 23.6295 N	1° 53' 56.8073 E	
F-11 IUH Guide 2 Target - plan hits target center - Point	0.00	360.00	2,953.00	-289.00	968.62	6,478,274.62	436,018.30	58° 26' 20.9819 N	1° 54' 14.9141 E	
F-11 IUH Guide 3 Target - plan hits target center - Point	0.00	0.00	3,004.70	-290.00	1,128.57	6,478,273.62	436,178.20	58° 26' 21.0337 N	1° 54' 24.7724 E	
F-11 IUH Guide 4 Target - plan hits target center - Point	0.00	360.00	3,079.50	-241.01	1,306.33	6,478,322.60	436,355.90	58° 26' 22.7103 N	1° 54' 35.6780 E	
F-11 IUH Guide 5 Target - plan hits target center - Point	0.00	360.00	3,132.00	-157.61	1,429.58	6,478,405.97	436,479.10	58° 26' 25.4698 N	1° 54' 43.1898 E	

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Survey Report

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Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B IOR DG3 rev0	Database:	Production EDM P246N

Casing Points					
	Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
	202.00	202.00	30"	30.000	36.000
	1,357.70	1,334.20	20"	20.000	26.000
	2,570.68	2,440.85	14"	14.000	17.500
	3,200.00	2,782.36	9 5/8"	9.625	12.250
	4,850.00	3,194.45	7"	7.000	8.500

Checked By: _____	Approved By: _____	Date: _____
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